

Member ID: _____

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NETWORK ADMINISTRATION USING CISCO® (315)

NATIONAL 2022

CONCEPT KNOWLEDGE:

Multiple Choice (50 @ 2 points each)

_____ (100 points)

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Contestant must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.

Multiple Choice: Identify the letter of the choice that *best* completes the statement or answers the question.

1. What port is commonly used for HTTP traffic?
 - a. 443
 - b. 80
 - c. 22
 - d. 8080
2. What command is used to clear all mac address table entries?
 - a. Clear mac-address-table
 - b. Clear macaddress table
 - c. Clean mac-address-table
 - d. Eliminate mac-address-table
3. Which of the following describes the AS number that is used in EIGRP?
 - a. Identifies the priority of the networks to be included in update packets.
 - b. Defines the reliability of the EIGRP routing protocol.
 - c. It is associated with the global Autonomous System number that is assigned by IANA
 - d. Specifies the EIGRP routing process on a router
4. Which dynamic routing protocol uses hop count to calculate metric value?
 - a. EIGRP
 - b. IS-IS
 - c. RIP
 - d. OSPF
5. What is one reason to use the *ip ospf priority* command when the OSPF routing protocol is in use?
 - a. Influence the DR/BDR election process
 - b. Provide a backdoor for connectivity during the convergence process
 - c. Activate the OSPF neighboring process
 - d. Streamline and speed up the convergence process
6. Which characteristic describes both ABRs and ASBRs that are implemented in a multiarea OSPF network?
 - a. They are required to reload frequently and quickly to update the LSDB
 - b. They both run multiple routing protocols simultaneously
 - c. They are required to perform any summarization or redistribution tasks
 - d. They usually have many local networks attached

7. What is used to facilitate hierarchical routing in OSPF?
 - a. Frequent SPF calculations
 - b. Auto summarization
 - c. The election of designated routers
 - d. The use of multiple areas
8. What is the purpose of a Null0 route in the routing table?
 - a. Prevent routing loops
 - b. Redistribute external routes into EIGRP
 - c. Act as a gateway of last resort
 - d. Prevent the router from sending EIGRP packets
9. What is the administrative distance of externally learned EIGRP routes?
 - a. 170
 - b. 120
 - c. 110
 - d. 90
10. When a Cisco router is configured with fast-switching, how are packets distributed over equal-cost paths?
 - a. Per-packet basis
 - b. Per-interface basis
 - c. Per-path-load basis
 - d. Per-destination basis
11. What capability is provided using application-specific integrated circuits in Cisco multilayer switches?
 - a. Aggregating physical ports into a single logical interface
 - b. Providing user authentication and authorization
 - c. Protecting data traffic in the event of a failed circuit
 - d. Forwarding of IP packets independent of the CPU
12. A network technician needs to connect a PC to a Cisco network device for initial configuration. What is required to perform this device configuration?
 - a. HTTP access
 - b. Telnet access
 - c. Internet access
 - d. Terminal emulation client
13. Which of the following statements describes a VPN?
 - a. They are open-source connections used to create tunnels over the internet
 - b. They have dedicated physical interfaces used by remote users
 - c. They use virtual connections to create a private network through a public network
 - d. They are logical connections to create a public network on the internet

14. What form of NAT is being used when inside addresses are mapped specifically to respective outside addresses?
- a. Static NAT
 - b. Dynamic NAT
 - c. PAT
 - d. Specific NAT
15. After configuring an access-list and applying it to a NAT configuration on a router at global config, the router still isn't performing NAT. What is most likely the issue?
- a. The NAT inside and outside interfaces have not been specified yet
 - b. NAT has not been enabled with the "IP NAT" command yet
 - c. NAT Table size has not been specified yet and there is no room to hold translations
 - d. NAT only works with named access control lists

16. A user enters the following command on a Cisco router:

```
access-list 110 permit 192.168.48.0 0.0.0.255
```

However, the user received an error instead. What is most likely the problem?

- a. The subnet is specified with a wild card mask. This instead needs to be a subnet mask.
- b. The ACL is created with a number greater than 99 which means this is an extended ACL. Extended ACLs need to have a protocol and a destination specified.
- c. There is no "any" keyword. This should be replaced with "all".
- d. The access control feature on the device has not been enabled. Enable it with the command "access-list enable".

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18. The following OSPF command was entered on a Cisco router:

```
network 192.168.0.0 0.0.255.255 area 0
```

However, IP address 192.168.1.2 on interface loopback 0 isn't being propagated through the network. What is most likely the issue?

- a. Interface loopback 0 is in a "down" state.
 - b. OSPF area 0 hasn't finished being configured on this router.
 - c. No adjacencies were formed on loopback 0 so its route isn't being distributed.
 - d. The OPSF setting for allowing loopback addresses has not been enabled.
19. A technician is troubleshooting a networking connectivity issue. As a troubleshooting step, she wants to clear the MAC address table on the Cisco switch. Which command will do this?
- a. Mac address-table clear
 - b. arp -d *
 - c. Reload mac-address-table
 - d. Clear mac-address-table

20. OSFP has been properly configured on a router and it is routing completely. However, when looking at the output of "show ip ospf neighbor", the following except is displayed

Neighbor ID	Pri	State	Dead Time	Address	Interface
192.168.1.2	1	2WAY/DROTHER	00:00:39	192.168.0.2	GigabitEthernet0/0
192.168.1.3	1	FULL/BDR	00:00:34	192.168.0.3	GigabitEthernet0/0
192.168.1.4	1	FULL/DR	00:00:32	192.168.0.4	GigabitEthernet0/0

Why is an adjacency not being formed with neighbor 192.168.1.2?

- a. It will not form an adjacency because only an adjacency with the designated router and backup designated router are needed.
 - b. The MTU on the interface of 192.168.1.2 does not match this router's interface.
 - c. OSPF has not been configured on 192.168.1.2 and therefore is not participating in OSPF.
 - d. Not enough time has passed for the adjacency to properly form.
21. How does a Cisco router act when it received two routes to the same network which are from different routing protocols and have the exact same cost?
- a. Both routes are put into the routing tables. The routes are used in a round-robin form to equally use both paths to the destination.
 - b. The route received on the lower number port is put into the routing table.
 - c. The route learned from the routing protocol with the lower administrative distance is placed into the routing table.
 - d. Both routes are put into the routing table. The second route is only used if the first route fails.

22. Starting with the network 192.168.128.0/24, how many different networks can be made by changing the subnet mask to 255.255.255.224?
- 8
 - 4
 - 16
 - 2
23. What is the correct wildcard mask to match the network 192.168.52.0/26?
- 0.0.0.63
 - 0.0.0.255
 - 0.0.0.31
 - 0.0.0.47
24. What is an appropriate subnet mask to support, as closely as possible, a classroom network of at least 34 student stations but not more than 60?
- 255.255.255.224
 - 255.255.255.192
 - 255.255.255.0
 - 255.255.255.240
25. A DHCP Server was configured with a network pool of 192.168.40.0/24 and default gateway of 192.168.40.1. Over 200 clients have verified that they have received IP addresses. However, only the first 127 clients can access other networks while the other half are not. What is most likely the issue?
- The DHCP server pool was setup wrong as a /24 network when the router's interface was setup as a /25 network.
 - The MAC Address table on the router became full and cannot accept more entries.
 - All the ethernet cables for the second half of the clients are bad.
 - All the ports for the second half of the clients are shutdown.
26. What command should be used to log messages so that there is a record when they are generated?
- NTP server 10.1.10.83
 - Clock timezone EST -5
 - Service timestamps log datetime
 - Show log time
27. When should an engineer measure network operation to create a baseline?
- Same time, every day to identify typical patterns
 - Different time, every day to identify abnormal patterns
 - Same day, different times to identify random patterns
 - During low work periods to identify minimum patterns

28. What is a function of the syslog service?
- a. Periodically poll information
 - b. Provide statistics of packets
 - c. Traffic analysis
 - d. Specify type of logging information captured
29. Which factor can be controlled when implementing QoS in a network?
- a. Delay
 - b. Speed
 - c. Routing
 - d. Addressing
30. What is a characteristic of voice traffic?
- a. It should not exceed 150ms
 - b. It requires at least 300kbs of bandwidth
 - c. It consumes a lot of resources
 - d. It is unpredictable
31. Why would a network engineer create a network baseline?
- a. Design the network
 - b. Select the equipment to deploy
 - c. Identify abnormal behavior
 - d. Evaluate security vulnerabilities
32. An engineer wants to deploy an EtherChannel configuration on a companies' network. Which of the following describes what they will be doing?
- a. Provide redundant links to support link failures
 - b. Provide redundant devices to support failures
 - c. Group multiple interfaces to increase bandwidth
 - d. Group devices to share a single interface
33. How would an engineer prevent packet loss on a congested interface?
- a. Disable queuing
 - b. Decrease buffer
 - c. Increase capacity on the link
 - d. Avoid traffic bursts
34. A network engineer does a ping test and receives several values in return. Which value shows the time it takes for a packet to travel to and from the destination?
- a. Latency
 - b. Speed
 - c. Delay
 - d. Priority

35. Which type of VPN includes using Digital certificates and public key infrastructure?

- a. IPsec
- b. GRE over IPsec
- c. SSL VPN
- d. Multipoint VPN

36. Which type of VPN is an enterprise-managed VPN?

- a. IPsec
- b. GRE tunnel
- c. Clientless SSL VPN
- d. IPsec VTI VPN

37. Which of the following is a public WAN infrastructure technology?

- a. VPN
- b. Cable
- c. Leased Lines
- d. DSL

38. Why would you use the overload keyword on the following command?

`ip nat inside source list 1 pool POOL overload`

- a. Allows many hosts to share an inside global address
- b. Allows a single host to use several global addresses
- c. Allows many external hosts to communicate with a single internal host
- d. Allow fast NAT translation by forcing table entries to be cached in the NAT ASIC

39. Which of the following is an advantage when deploying NAT for IPv4 in a network?

- a. Improving packet handling
- b. Authentication capability on IPv4
- c. Slows IPv4 depletion
- d. Impacts switch performance

40. What does the output of “show access-lists” display?

- a. All access-lists that have been defined on the device
- b. All access-lists that have been defined and are currently being used on the device
- c. Only access-lists that are currently being used on the device
- d. Only named access-lists

41. Reading the following output, what is the effect you should expect?

```
Market-SW11# show interface switchport
...
Name: Fa0/24
Switchport: Enabled
Administrative Mode: trunk
Operational Mode: trunk
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: dot1q
Negotiation of Trunking: On
Access Mode VLAN: 1 (default)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: none
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: 1-29,31-1005
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
Unknown unicast blocked: disabled
Unknown multicast blocked: disabled
Appliance trust: none
```

- a. The port is in trunk mode and is trunking for all VLANs except VLAN 30
 - b. The port will dynamically become a trunk port if it detects if the remote port is in trunk mode
 - c. Switchport operations on this port have been disabled
 - d. VLAN 10 is the native VLAN for this port
42. When a port on a Cisco switch is configured to be a trunk port, what is the default behavior for VLANs?
- a. Only VLAN 1 is trunked by default
 - b. VLANs lower than 100 are trunked by default
 - c. VLANs over 100 are trunked by default
 - d. All VLANs are trunked by default

43. What is the purpose of the “native” VLAN for a trunk port?
- It is the specified VLAN that the trunk port will not tag frames when sending and any frames received without tags will be directed to that VLAN.
 - It is another name for the Default VLAN.
 - It is the first or primary VLAN that the trunk port will attempt to send frames with VLAN tags.
 - It is a required setting to allow all communication with a Cisco Router.
44. While you are connected to the console port of a Cisco switch, you keep receiving the following error:
- `%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on FastEthernet0/23 (1), with SW11 FastEthernet0/1 (10).`
- What is most likely the issue?
- The native VLAN setting for the local port Fa0/23 is set to VLAN 10 while on the remote switch’s port it is set to VLAN 1.
 - The native VLAN setting for the local port Fa0/23 is set to VLAN 1 while on the remote switch’s port it is set to VLAN 10.
 - CDP has been improperly configured to share the native VLAN value.
 - The local port Fa0/23 was not able to learn the appropriate Native VLAN setting from the remote port.
45. A technician is setting up a new server that will be used for many different services. One of these services is an SSH server. However, to protect it from hacking, the technician changes the port it uses to port 443. Why might this be a bad idea?
- Port 443 is commonly used for HTTPS traffic. This could disrupt other services running on the server
 - Port 443 is out of the range of usable TCP/UDP ports
 - SSH cannot run on port 443 as it can only operate on port 22
 - Manual port overrides can only be assigned to TCP/UDP ports greater than 10,000
46. A client describes that they have software that communicates with other computers running the same software. The software does have a setting to state what transport protocol is used in the transmission of data. For the client, it is important that the data is transmitted successfully and there is no data loss. If there are lost packets or errors in the transmission, corrective actions should be taken to ensure proper data delivery. What should the client choose for the transport protocol?
- UDP
 - HTTP
 - TCP
 - NTP

47. After setting an IP address on a switch, a technician is not able to ping or remote into it with a direct ethernet connection to the switch. What is most likely the issue?
- a. The technician set the IP address on VLAN 1 but forgot to run the command “no shutdown”.
 - b. The wrong user account was used to set the IP address.
 - c. The default gateway for the switch was set to the wrong IP address.
 - d. The default firewall on the switch is enabled and blocking ping requests.
48. After working for hours configuring a Cisco switch, the technician left for the night. During the night, there was a short power outage. In the morning it was found that the switch had some but not all its configuration. What most likely happened?
- a. The switch configuration periodically backups up to a network location but had not finished the full backup by the time the power outage occurred.
 - b. After the power outage, the switch booted into its backup partition where the configuration cannot be reached. The switch just needs to be rebooted again to access the configuration once again.
 - c. The configuration does not automatically load when the switch is booted. The network administrator needs to reapply the configuration that is stored in NVRAM.
 - d. The technician saved the configuration at some point during configuration but forgot to save the final running configuration.
49. Review the following output.

```
Corp-A-R1(conf)#
```

What mode are you currently in?

- a. Global configuration mode
 - b. Privileged exec mode
 - c. User exec mode
 - d. Interface sub-configuration mode
50. Which of the following is a layer-2 protocol?
- a. Point-to-point protocol
 - b. IPX/SPX
 - c. HTTPS
 - d. Fiber optic